AgroLnk Exchange

End-to-End Operational Guide: Auctions, Escrow Payouts, & Logistics

OFFICIAL ARCHITECTURE REFERENCE

Release 2026.1 • Escrow Trust & Ticker Engine

1. End-to-End Workflow at a Glance

AgroLnk connects agricultural producers and food processors through a 100% verified, escrow-secured trading infrastructure. The diagram below illustrates how an auction transforms into a settled financial contract and physical delivery:

1
Auction Won
Timer expires or farmer accepts early bid. Reserve price verified.

2
Order Generated
Order #AGM-XXXX auto-created. Escrow status: *pending_deposit*.

3
2-OTP Transport
Transporter assigned. Farmgate Pickup OTP & Destination Delivery OTP.

4
Escrow Release
Buyer confirms arrival & quality. Funds disbursed directly to farmer.

2. Next Steps After Winning an Auction

> **Key Rule:** When the auction timer reaches 00:00:00 (or the farmer clicks "*Accept Offer & Close Now*"), the system executes an atomic transaction (`finalize_auction_atomic`). If the winning bid meets or exceeds the farmer's **Reserve Floor Price**, the auction is formally completed and the sale contract is bound.

Action Items for the Winning Buyer:

- **Instant Victory Notification:** A trophy modal banner appears in the live auction room declaring the winning rate and total valuation.
- **Order Redirect:** The buyer clicks `View Order` or navigates to `Orders & Dispatch -> Purchases`.
- **Escrow Deposit (Locking Funds):** The buyer deposits the purchase value into AgroLnk's Bank Escrow Trust Account. This guarantees that farmer will get paid immediately upon delivery.
- **Delivery Destination Confirmation:** The buyer verifies the destination warehouse address and contact details for incoming freight.

Action Items for the Seller / Farmer:

- **Order Created Notification:** The farmer receives an alert in their Notification Bell: "*Auction Won! Order #AGM-XXXX created. Awaiting buyer escrow funding.*"
- **Lot Preparation:** Farmer bags and prepares the harvest lot at farmgate or aggregation godown.
- **Pickup Security Code (OTP):** A secure 4-digit Pickup OTP is generated in the farmer's portal to verify the transporter before allowing truck loading.

3. How Notifications and Alerts Pass Between Parties

AgroLnk employs a dual-channel notification architecture:

- **Real-Time WebSocket Feed (Supabase Realtime):** When a new bid is placed, ticker prices update instantly across all active screens with sub-second latency.
- **State-Driven User Alerts (In-App Bell & Storage Sync):** Targeted alerts delivered strictly to relevant roles without spam.

Event Trigger	Recipient	Alert Channel	Actionable Outcome
Outbid Event	Previous Leading Bidder	Screen Banner + Sound	Alerts user they are outbid; allows instant counter-bid.
Early Knockdown	All Bidders in Room	Live Ticker Broadcast	Freezes timer; declares winner chosen by farmer.
Auction Won	Winner (Buyer) & Seller	In-App Bell + Modal	Provides 1-click button to Escrow Order checkout.
Escrow Funded	Farmer & Transporter	Notification Bell + Badge	Authorizes farmer to release produce to the truck.
Consignment Dispatched	Buyer	Live Delivery Tracker	Displays vehicle number, driver phone, and live route.
Arrival & Inspection	Escrow Desk & Farmer	Order Detail Badge	Triggers escrow release to farmer's verified bank account.

4. Is the "Order Release" / Payment Release Option Available?

> **YES, 100% AVAILABLE:** AgroLnk utilizes a strict digital escrow trust protocol to protect both farmers from default and buyers from fraudulent or sub-standard consignments.

How Escrow Release Works Step-by-Step:

1. **Escrow Funded (Deposit Held in Trust):**
   The buyer deposits the total contract amount. Funds are locked securely under bank custody (IDBI / Razorpay Escrow Trust). The farmer sees the status `Protected in Escrow`.
2. **Physical Delivery & Inspection:**
   The transport truck arrives at the buyer's destination mill or depot. The buyer inspects the consignment for quantity, moisture, and grade compliance.
3. **Delivery OTP Verification:**
   The buyer provides the 4-digit Delivery OTP to the driver, confirming physical handover.
4. **Release Trigger (Buyer Confirmation & Ops Verification):**
   The buyer clicks `Confirm Order Receipt` in their portal. In accordance with APMC guidelines, AgroLnk Operations desk conducts an automated / quick telephone audit to verify assay parameters.
5. **Disbursement to Farmer Bank Account:**
   Funds are immediately disbursed via IMPS / NEFT directly into the farmer's verified bank account. A **Bank UTR Reference Number** (e.g. `UTR-20261030-98421`) is stamped permanently on the order invoice. The order changes to `Settled to Bank (Completed)`.

5. How the Logistics & Transport Process Works

When an auction completes and generates an order, AgroLnk automatically instantiates a companion delivery consignment (`DEL-XXXX`) in the logistics system.

Logistics Execution Sequence:

A. Automatic Vehicle & Tariff Calculation

Distance is computed using our district transit matrix. Freight is priced according to transparent vehicle categories:

- **Mini Truck (Tata Ace / Bolero):** Base ₹400 + ₹14/km (lots up to 1.5 MT)
- **Medium LCV (14ft Eicher):** Base ₹900 + ₹24/km (standard loads up to 5.0 MT)
- **Reefer Cold-Chain Truck:** Base ₹1600 + ₹34/km (temperature-controlled perishables)

B. Transporter Assignment

Verified logistics partners in the origin cluster review the delivery request via their `Transporter Dashboard`. Once assigned, driver name, phone number, and vehicle registration are visible to both buyer and farmer.

C. Two-Step Anti-Theft OTP Protocol

Step 1 (Pickup OTP): When the truck reaches the farmgate, the farmer inspects the truck and shares the *Pickup OTP* with the driver. This moves the order to `In Transit`.
Step 2 (Delivery OTP): Upon safe arrival at the buyer facility, the buyer gives the *Delivery OTP* to confirm unloading.

6. Buyer Confirmation Expiry, Farmer Decisions & Runner-Up Policy

> **Payment & Confirmation Window: 24 Hours Standard**
> The winning bidder has a strict **24-hour window** to fund the escrow trust account. During this period, the farmer's lot is reserved exclusively for the winning buyer.

Scenario A: The Winning Buyer Confirms & Funds Escrow

The contract proceeds smoothly into pickup, dispatch, and final settlement.

Scenario B: The Winning Buyer Fails to Confirm (Expires or Backs Out)

If the 24-hour window lapses without payment deposit, the winning bid is declared **defaulted**. The farmer has two powerful, one-click options:

1. **Offer to Runner-Up (2nd Highest Bidder):**
   The farmer can instantly extend a purchase offer to the 2nd highest bidder at their recorded bid price. The 2nd bidder receives an urgent alert: "*Opportunity: The leading lot is offered to you at ₹X/unit. Accept within 12 hours to claim.*"
2. **1-Click Relist / Re-Auction:**
   If the runner-up offer is not desired, the farmer clicks `Relist Produce`. The system automatically pre-populates all commodity details, photos, and grades, allowing the farmer to launch a fresh 30-minute or 2-hour live auction with an updated reserve price.

7. Frequently Asked Operational Questions

Q1: Can a farmer bid on their own auction to artificially raise the price?

No. AgroLnk's database RPC enforces strict identity isolation. Producers are forbidden from bidding on their own lots, and the leading bidder cannot place consecutive bids against themselves.

Q2: What happens if the highest bid is below the farmer's Reserve Floor?

The auction status changes to `reserve_not_met`. No order is created, and the farmer is under no legal obligation to sell. The farmer can freely relist or offer the lot in the general produce marketplace.

Q3: What if the buyer finds quality defects upon truck arrival?

AgroLnk includes an in-order `Inspect Assay Quality` modal. If the lot deviates from the declared grade (e.g. moisture > 14% or foreign matter), the buyer can raise a formal dispute before releasing escrow. An AgroLnk certified quality surveyor reviews the assay data to negotiate an equitable price adjustment or return freight.

Q4: How does a Reverse Auction differ from a Standard Forward Auction?

In a **Forward Auction**, buyers compete by submitting increasingly higher bids (₹+25 step). In a **Reverse Auction** (procurement contract), suppliers compete by submitting lower bids down to the buyer's reserve ceiling. Both models utilize identical escrow, OTP, and dispatch mechanics.